

PRATHAM SHARMA

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EDUCATION

Bachelor of Technology (B.Tech) in Electronics Instrumentation and Control

Thapar Institute Of Engineering And Technology

| 08/2023 - 05/2027 | Patiala

Minor in Computer Science & Engineering

Thapar Institute Of Engineering And Technology

| 08/2024 - 05/2026 | Patiala

CERTIFICATIONS

Bootcamp On DFTA : Drone Flight Technology

Bootcamp covering Drone Flight Technology under the aegis of SwaYaan and MeitY

Python For Data Science

Python programming with data science applications by MSME - Technology center

FDP Quantum Machine Learning

Certification in quantum ML : theory, Implementation and practical Application

Certified IOS And Android Developer

Certification for developing applications on iOS and Android platforms by WhiteHat Jr

Certified Game Developer

Certification in game development by WhiteHat Jr

Certified UI/UX Designer

Certification in user interface and user experience design by WhiteHat Jr

SKILLS

Programming & Data:

Quantum Machine Learning, Python, C/C++, Java, R Programming, Data Structures, OOPS, SQL, PL/SQL, MongoDB

Hardware & Embedded:

ARM Microcontroller, FPGA, STM32 (HAL, CubeMX), ESP32, Arduino, Raspberry Pi, Embedded Systems, Instrumentation, IOT

Robotics & AI:

ROS, Robotics, Computer Vision, Deep Learning, Open CV, Machine Learning, CNN, MATLAB

Design & Development:

JavaScript, Node.js, React, HTML, EAGLE, AutoCAD, UI/UX Design, Game Development, Mobile Application Development

Systems & Networking:

Operating Systems, Database Management Systems (DBMS), Computer Networking, CISCO Packet tracer

KEY ACHIEVEMENTS

Received “Award of Honor” at Bootcamp On DFTA :

Scored highest at **Bootcamp On DFTA : Drone Flight Technology And Application**

PROJECTS

XAI Autonomous Rover and Robotic Arm for Disease and Pest Detection In Crops

 06/2025  Thapar Institute Of Engineering And Technology

Created an XAI Autonomous Rover capable of diagnosing crop health issues

- Developed an autonomous rover for crop disease and pest detection
- Integrated **ROS2**, **Raspberry Pi 5**, and **OpenCV** for pixel-to-world coordinate mapping.
- Utilized machine learning and deep learning for image analysis
- Employed advanced techniques like YOLO and RESNET for enhanced detection

Drone Flight Technology Bootcamp | SwaYaan & MeitY

 08/2025  Thapar Institute Of Engineering And Technology

Under the aegis of Ministry of Electronics and Information Technology and SwaYaan

- Worked on quadcopter drone architecture, flight stabilization, and autonomous flight concepts using Pixhawk flight controller.
- Studied IMU-based sensor fusion using gyroscope, accelerometer, magnetometer, and GPS data for attitude estimation and navigation.
- Implemented and analysed PID-based flight stabilization for roll, pitch, and yaw control.
- Configured ESC–BLDC motor interfacing, RC communication, telemetry systems, and flight calibration procedures.
- Learned autonomous flight operations including waypoint navigation, altitude hold, and return-to-home functionality. Also studied DGCA drone safety regulations, UAV classifications, and no-fly zone compliance.

ROBOTIC ARM & MATLAB simulation

 05/2025  Thapar Institute Of Engineering And Technology

Developed a robotic arm and simulation project using advanced algorithms

- Developed a robotic arm simulation using MATLAB
- Utilized advanced programming for robotic control

IOT based Home Automation

 02/2025  Thapar Institute Of Engineering And Technology

Developed an IoT system for home automation to enhance user convenience

- Created an IoT-based home automation system
- Implemented user-friendly interfaces for smart home management

Smart Home Security System Using Arduino

 07/2024  Thapar Institute Of Engineering And Technology

Developed a smart home security solution utilizing Arduino technology

- Built a smart home security system using Arduino
- Ensured reliable detection and notification of potential security threats

Arduino based Embedded system

 08/2023  Thapar Institute Of Engineering And Technology

Created an embedded system project based on Arduino for enhanced functionality

- Designed and created an Arduino-based embedded system project
- Focused on real-time monitoring and control